

CampusEats — Assignment 3

Integrate an External SOAP Partner

Team ID: CampusEats-MultiAgent-01

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1. Context

CampusEats integrates an external **Payment Gateway** using the single operation **charge**. The rest of CampusEats can remain REST-based, while this financial integration is modeled with SOAP because SOAP provides a strong WSDL contract, structured faults and a clear message-oriented security boundary. This is design-time only; no live gateway or server is implemented.

2. WSDL Contract

The supplied partner.wsdl contains types, messages, portType, SOAP-over-HTTP binding, service and port. It defines the charge operation.

Element	Definition
Operation	charge
Input	merchantId, orderId, amount, currency, paymentToken
Output	status, transactionId, orderId, message
SOAPAction	http://campuseats.example.com/payment/charge
Endpoint	https://payments.partner.example.com/soap/PaymentGateway
Binding	document/literal SOAP over HTTP

3. SOAP Messages

Request — credentials are in Header and charge is in Body.

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:pay="http://campuseats.example.com/payment">
  <soap:Header>
    <pay:Credentials>
      <pay:merchantId>CE-CAMPUS-001</pay:merchantId>
      <pay:apiKey>REDACTED-DEMO-KEY</pay:apiKey>
      <pay:timestamp>2026-08-26T22:30:00+05:30</pay:timestamp>
    </pay:Credentials>
  </soap:Header>
  <soap:Body>
    <pay:ChargeRequest>
      <pay:merchantId>CE-CAMPUS-001</pay:merchantId>
      <pay:orderId>CE-ORD-10045</pay:orderId>
      <pay:amount>249.00</pay:amount>
      <pay:currency>INR</pay:currency>
      <pay:paymentToken>tok_demo_7F83A2</pay:paymentToken>
    </pay:ChargeRequest>
  </soap:Body>
</soap:Envelope>
```

Success response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:pay="http://campuseats.example.com/payment">
  <soap:Header/>
  <soap:Body>
    <pay:ChargeResponse>
      <pay:status>APPROVED</pay:status>
      <pay:transactionId>TXN-20260826-88421</pay:transactionId>
      <pay:orderId>CE-ORD-10045</pay:orderId>
      <pay:message>Payment approved</pay:message>
    </pay:ChargeResponse>
  </soap:Body>
</soap:Envelope>
```

Fault response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Header/>
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:Client</faultcode>
      <faultstring>Payment declined</faultstring>
      <detail><PaymentFault>
        <code>card_declined</code>
        <orderId>CE-ORD-10045</orderId>
        <message>Issuer declined the payment.</message>
      </PaymentFault></detail>
    </soap:Fault>
```

```
</soap:Body>
</soap:Envelope>
```

4. HTTP Binding

```
POST /soap/PaymentGateway HTTP/1.1
Host: payments.partner.example.com
Content-Type: text/xml; charset=utf-8
SOAPAction: "http://campuseats.example.com/payment/charge"
```

```
https://payments.partner.example.com/soap/PaymentGateway
```

```
<soap:Envelope> ... </soap:Envelope>
```

5. Discovery

CampusEats discovers the partner through a modern catalogue/registry entry. No UDDI server is required. The catalogue points the consumer to the partner WSDL.

Field	Registry entry
business	CampusPay External Payment Services
service	CampusEats Payment Gateway
endpoint	https://payments.partner.example.com/soap/PaymentGateway
WSDL / tModel-style pointer	https://payments.partner.example.com/wsdl/PaymentGateway.wsdl
operation	charge

6. Fault Mapping

The partner fault **card_declined** is mapped internally to the CampusEats contract error **placeOrder** → **payment declined**. The partner vocabulary is not exposed to students. The student receives a stable message such as: **Payment could not be completed for this order.**